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B.Tech. Degree IV Semester Examination April 2019

ME 1405 HYDRAULIC MACHINERY (2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) What is meant by geometric, kinematic and dynamic similarities?
- (b) Write a note on Rayleigh's method of dimensional analysis.
- (c) What do you mean by specific speed of a turbine?
- (d) What is the importance of draft tube in reaction turbines?
- (e) What do you mean by NPSH of a centrifugal pump?
- (f) Describe the use of air vessels in a reciprocating pump.
- (g) Explain the working of a hydraulic coupling.
- (h) Explain the working of a gear pump with neat sketch.

PART B

(4 × 15 = 60)

- II. (a) Using the method of dimensional analysis obtain an expression for the discharge Q over a rectangular weir. The discharge depends on the head H over the weir, acceleration due to gravity g , length of weir crest L , height of the weir crest over the channel bottom Z and the kinematic viscosity γ of the liquid. (10)
- (b) A geometrically similar model of spillway is to be laid to a scale of 1 in 50. Calculate the velocity ratio, discharge ratio and acceleration ratio. (5)

OR

- III. A 1/40 model of an ocean tanker is dragged through fresh water at 2 m/s with a total measured drag of 120 N. The skin (friction) drag co-efficient ' f ' for model and prototype are 0.3 and 0.02 respectively in the equation $R_f = fAV^2$. The wetted surface area of the model is 25 m². Taking the densities for the prototype and model as 1030 kg/m³ and 1000 kg/m³ respectively, determine: (i) the total drag on the prototype (ii) power required to drive the prototype. (15)

- IV. (a) Write a note on axial flow reaction turbine with a neat sketch. (6)
- (b) A Kaplan turbine runner is to be designed to develop 9000 kW shaft power. The net available head is 6 m. Assume that the speed ratio is 2.5, flow ratio is 0.7, and the overall efficiency is 65%. The diameter of the runner is 4 times the boss diameter. Find the diameter and speed of the runner and specific speed of the turbine. (9)

OR

- V. (a) Write a note on governing of impulse turbine. (7)
- (b) A Pelton wheel has to be designed for the following data: Power to be developed = 6 MW. Net head available = 300 m; ratio of jet diameter to wheel diameter = 1/10; speed = 550 rpm and overall efficiency = 85%. Find the number of jets required, diameter of jet, diameter of wheel and quantity of water required (assume $C_v = 0.98$, speed ratio, $K_u = 0.46$). (8)

(P.T.O.)

- VI. The impeller of centrifugal pump has an external diameter of 450 mm and internal diameter of 250 mm and it runs at 1500 rpm. Assuming a constant radial flow through the impeller at 3 m/s and that at exit set back at an angle of 28° , draw the velocity triangle and determine inlet vane angle, the angle made by the absolute velocity at outlet with the direction of impeller motion and the work done by the impeller on water. (15)

OR

- VII. (a) Write a note on multistage centrifugal pumps with neat sketches. (5)
(b) A single acting reciprocating pump has a piston diameter of 150 mm and stroke length 300 mm. The length and diameter of the suction pipe are 8 m and 60 mm respectively. If the suction lift of the pump is 3.2 m and separation occurs when pressure in the pump falls below 2.5 m of water absolute and the manometer reads 765 mm of mercury, find the maximum speed at which pump can be run without separation in the suction pipe. (10)

- VIII. Explain the working of Hydraulic accumulator, Hydraulic ram and Vane pump with neat sketches. (15)

OR

- IX. Write notes on (i) Hydraulic intensifier (ii) Hydraulic press (iii) Surge tank. (15)
